

Signal Scaling and Calibration

WTS-100 Water Treatment and Transfer Skid

How a sensor reading becomes a number the PLC can use, and how to tell when that number has gone bad.

The block FB_AIN_SCALE in the PLC program implements this document.

1. Why 4 to 20 mA

Transmitters send a current, not a voltage. 4 mA is the bottom of the range, 20 mA is the top.

Cable length does not matter. In a series loop the current is the same at every point. A voltage signal drops across the resistance of a long cable and reads low at the far end.

A broken wire is detectable. Empty reads 4 mA, not 0 mA. So 0 mA can only mean something is broken. If the signal were 0 to 10 V, a cut wire would read 0 V, which looks exactly like a real zero reading. You would never know.

This is called a live zero.

Loop voltage check, worst case

Item	Volts
Transmitter minimum operating voltage	12.0
Sense resistor 250 ohm at 20 mA	5.0
Barrier and terminal drops	1.5
Cable, 2 x 100 m at 0.09 ohm/m, 20 mA	0.4
Total needed	18.9
Supply	24.0
Spare	5.1

Passes with margin.

2. Counts

The analog input card turns the current into a whole number.

Current	Counts	Meaning
below 3.6 mA	below -691	broken wire or dead transmitter
4.000 mA	0	0 percent
12.000 mA	13824	50 percent
20.000 mA	27648	100 percent
above 20.5 mA	above 28512	transmitter failed high or short circuit

There are 1728 counts per milliamp.

The formula

```
value in engineering units = EU_min + (counts / 27648) x (EU_max - EU_min)
```

For an analog output, the same thing backwards:

```
counts = (value - EU_min) / (EU_max - EU_min) x 27648
```

3. Scaling for each instrument

Tag	Range	Counts per unit	Smallest step
LT-101	0 to 3000 mm	9.216	0.109 mm
LT-102	0 to 2500 mm	11.059	0.090 mm
FT-101	0 to 50 m ³ /h	552.96	0.0018 m ³ /h
FT-103	0 to 500 L/h	55.296	0.018 L/h
TT-102	0 to 100 degC	276.48	0.0036 degC
PT-101	0 to 10 bar	2764.8	0.0004 bar
AT-102	0 to 2000 uS/cm	13.824	0.072 uS/cm

The card can resolve far finer than any of these sensors can measure. That is the right way round. The measurement is limited by the instrument, not by the card.

4. How accurate the reading actually is

Errors from different sources are independent, so they are combined as a root sum of squares rather than simply added.

```
total = square root of ( sensor^2 + transmitter^2 + card^2 + drift^2 )
```

Temperature, TT-102

Source	Error
Pt100 Class A sensor at 50 degC	0.25 degC
Head transmitter	0.10 degC
Analog card	0.007 degC
Ambient drift over a 20 degC swing	0.20 degC
Combined	0.34 degC

This is why the temperature deadband is 0.5 degC. The controller cannot tell the difference between a real change and measurement noise below about 0.34 degC. A tighter deadband would make it react to noise instead of to the process. 0.5 degC sits just outside the uncertainty, at about 1.5 times it.

The deadband was calculated, not chosen.

Flow, FT-101

Source	Error
Magnetic flowmeter, 0.5 percent of reading	0.25 m ³ /h at full flow
Transmitter	0.05 m ³ /h
Analog card	0.004 m ³ /h
Combined	0.255 m³/h, or 0.51 percent of span

Note this is a percentage of the reading, not of the range. At 10 m³/h the absolute error falls to about 0.05 m³/h. Below 1.0 m³/h the reading is mostly noise, so it is forced to zero.

5. When a reading goes bad

Problem	How it is spotted	What the PLC does
Broken wire	counts below -691	mark invalid, hold the last good value, alarm
Transmitter failed high	counts above 28512	mark invalid, clamp to full scale, alarm
Reading frozen	value has not moved at all for 5 minutes while the process should be changing	alarm only, do not trip

Why hold the last good value instead of dropping to zero. A control loop fed a sudden zero would slam its output to the limit. Holding the last good value keeps the plant steady and gives the sequence 30 seconds to shut down in a controlled way. That is interlock I7 in the control narrative.

Why a frozen reading only alarms. A steady process legitimately produces a steady reading. A frozen signal is a reason to go and look, not a reason to trip.

A frozen sensor is the dangerous case, because the reading stays inside the valid range and looks perfectly healthy. Range checks alone will never catch it.

6. Calibration

Done at commissioning and once a year. Each loop is calibrated as a whole loop: inject at the transmitter, read the value on the HMI. Not at the card. That way a wiring fault and a scaling fault are both caught.

For each analog input:

1. Isolate the instrument and connect a calibrated current source in its place.
2. Inject 4, 8, 12, 16 and 20 mA in turn. That is 0, 25, 50, 75 and 100 percent.
3. Write down what the HMI shows at each point.
4. Work out the error as a percentage of span. It must be inside the figure

from section 4.

1. Repeat going back down, 20 to 4 mA, to catch hysteresis.
2. Record the results. A loop that fails is tagged and not commissioned.

Record both as found and as left. As found tells you whether the loop has drifted since last time. As left proves it is correct now.

Zero and span

Zero moves the whole line up or down. Adjust it at 4 mA.

Span changes the slope of the line. Adjust it at 20 mA.

They interact, so always go back and check zero after adjusting span, and repeat until both are right.